

Complete Categorized Interview Question List

1. Core Java

- What is a Sealed Class in Java?
 - Difference between Final Class and Sealed Class
 - What is Record in Java?
 - Java memory model segments
 - Heap vs Stack vs other memory areas
 - What is Reflection?
 - What is Reflection API?
 - What is Garbage Collection?
 - Thread lifecycle
 - Difference between start() and run()
 - Can a thread be restarted?
 - What is InterruptedException?
 - What is volatile keyword?
 - What is AtomicInteger?
 - How CAS works?
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2. Java 8 / Functional Programming

- Reverse string using Streams
 - Java streams intermediate vs terminal operations
 - Difference between Collection vs Streams
 - Infinite streams
 - Optional class
 - Optional.isPresent() uses which functional interface?
 - Difference between map() and flatMap()
 - What are terminal operations?
 - Is forEach terminal?
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3. Collections

- Types of Java Collections
- Hash collision
- How HashMap handles collisions

- ConcurrentHashMap internals
 - HashSet implementation
 - Difference between HashMap and HashSet
 - What happens if same key inserted twice
 - What happens when HashMap exceeds 16 elements
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4. Concurrency / Multithreading

- Ways to create threads
 - ExecutorService
 - CompletableFuture
 - How to return value from CompletableFuture
 - Chaining async APIs
 - synchronized keyword
 - ThreadLocal class
 - AtomicLong vs LongAdder
 - Thread pool usage
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5. Design Patterns

- What are design patterns?
 - Factory pattern
 - Singleton pattern
 - Making Singleton thread safe
 - Singleton issues
 - Reflection
 - Serialization
 - Cloning
 - Decorator pattern
 - SOLID principles
 - Liskov Substitution Principle
 - Open Closed Principle
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6. Spring Boot

- What is @Qualifier
- Auto configuration
- Autowiring
- Custom endpoints
- Custom query in Spring Data JPA
- Global exception handling

- REST controller structure
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7. Microservices

- API Gateway
 - Service authentication
 - Circuit breaker pattern
 - Inter-service communication
 - Retry strategies
 - Dead Letter Queue
 - Idempotency
 - Exactly-once processing
 - Saga pattern
 - CQRS
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8. REST APIs

- What is REST?
 - REST best practices
 - API pagination
 - HTTP status codes
 - HTTP 202 meaning
 - GET vs POST same path
 - Request JSON response XML
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9. Security

- Stateless JWT
 - Session tokens
 - Spring Security annotations
 - CSRF
 - API security approaches
 - Password handling
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10 Database

- ACID properties

- Isolation levels
 - MongoDB indexing
 - MongoDB transactions
 - Removing duplicate records
 - N+1 problem
 - Lazy vs Eager fetching
 - Optimistic vs pessimistic locking
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11. Coding Questions

- Reverse string
 - Reverse string with special characters
 - First non-repeating character
 - Anagram program
 - Count word occurrences
 - Prime numbers using Java 8
 - Group by department & average salary
 - Sort RGB characters
 - LRU cache implementation
 - LeetCode(Medium & Easy) – sliding window
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12. System Design

- Design Tic Tac Toe
 - Browser back/forward navigation
 - Highly concurrent counter
 - System responding within 3 seconds
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13. Cloud / DevOps

- CI/CD pipeline
 - Blue-green deployment
 - Canary deployment
 - High availability
 - Disaster recovery
 - GCP deployment
 - Terraform vs GCP
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14. Linux / Networking

- grep command
 - cat command
 - Check processes using most resources
 - TCP vs UDP
 - Connection vs connectionless
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15. Testing

- Unit test vs integration test
 - JUnit lifecycle
 - Mockito
 - @Mock vs @MockBean
 - PowerMock
 - Test Driven Development
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16. Agile / Process

- Jira workflow
- User stories
- Story points
- Sprint
- Program Increment